

Logistic Regression

Logistic Regression will teach us how to build predictive models when the outcome is categorical rather than continuous. Instead of predicting a numerical value, it estimates the probability that an observation belongs to a particular class, such as whether a borrower is a good or bad credit risk, whether a recession may occur, or whether the stock market may move up or down. This makes Logistic Regression useful for classification problems in finance and economics. It also helps us understand important evaluation tools such as confusion matrices, precision, recall, F1-score, ROC curves and classification thresholds.

Problem	Overview	Research-paper anchor	Dataset
Problem 5	Predict good vs bad credit risk	Credit scoring literature using German Credit	UCI Statlog German Credit
Problem 6	Predict U.S. recession probability	Estrella & Mishkin yield-curve recession-prediction literature; the Fed note discusses recession probability models using term spreads. (<u>Federal Reserve Bank of New York</u>)	FRED: 10-year Treasury, 3-month Treasury bill, NBER recession indicator
Problem 7	Forecast monthly stock-market direction	Nyberg 2011 forecasts the sign of monthly excess stock returns using dynamic binary models. (<u>EconPapers</u>)	Kenneth French market excess return + FRED/FRED-MD macro predictors

Table 1: Overview of Problems, Research Anchors, and Datasets

5. Credit Risk Classification Using German Credit Data

Problem 5: Logistic Regression – Credit Risk Classification Using German Credit Data

Research Anchor: Beninel, Bouaguel and Belmufti (2012), “Transfer Learning Using Logistic Regression in Credit Scoring”

This problem introduces Logistic Regression as the first classification model in Phase 1. Unlike Linear Regression, which predicts a continuous numerical value, Logistic Regression is used when the outcome variable is categorical.

In this exercise, the objective is to predict **whether a borrower is a good or bad credit risk** using the German Credit dataset. The target variable is converted into a binary classification problem, where 0 represents good credit risk and 1 represents bad credit risk. The model is evaluated using accuracy, confusion matrix, classification report, ROC-AUC score, and the ROC curve.

Dataset: Statlog German Credit Data

Dataset Source: UCI Machine Learning Repository

The German Credit dataset from the UCI Machine Learning Repository classifies people as good or bad credit risks and includes a cost matrix.

Logistic Regression predicts a probability. Example: Probability of bad credit risk = 0.73. Then we convert that probability into a class:

Probability of bad risk	Predicted class
Below 0.50	Good credit risk
0.50 or above	Bad credit risk

Table 2: Borrower Classification

So, the question for this problem is: Can we use borrower characteristics to predict whether a person is a good or bad credit risk?

In this dataset:

- 1 = Good credit risk
- 2 = Bad credit risk

But for modelling, it is easier to define:

- 0 = Good credit risk
- 1 = Bad credit risk

X = borrower characteristics

y = whether the borrower is bad credit risk

In credit risk, accuracy alone is not enough. We also need the confusion matrix.

	Predicted Good	Predicted Bad
Actually Good	True Negative	False Positive
Actually Bad	False Negative	True Positive

Table 3: Confusion Matrix for Credit Risk Classification

The most dangerous mistake in credit risk is usually: Actually a bad borrower, but predicted as good.

Metric	Meaning
Precision	Of those predicted bad, how many were actually bad?
Recall	Of actual bad borrowers, how many did we catch?
F1-score	Balance between precision and recall

Table 4: Classification Report Layout

For credit risk, **recall for bad borrowers** is very important. ROC-AUC measures how well the model separates good and bad borrowers.

AUC	Interpretation
0.50	No better than random
0.60–0.70	Weak/moderate
0.70–0.80	Reasonable

0.80+ Strong

Table 5: ROC - AUC Interpretation**Findings**

Metric	Value
Accuracy	0.77
ROC-AUC	0.796
Test observations	300

Table 6: Main Results

The model correctly classified about **77%** of borrowers in the test set.

The ROC-AUC score is: **0.796**

This is fairly good for a beginner credit-risk model. It means: The model has a reasonable ability to separate good-risk borrowers from bad-risk borrowers.

	Predicted Good	Predicted Bad
Actually Good	185	25
Actually Bad	44	46

Table 7: Confusion Matrix Results

Result	Count	Meaning
True Negative	185	Good borrowers correctly predicted as good
False Positive	25	Good borrowers wrongly predicted as bad
False Negative	44	Bad borrowers wrongly predicted as good
True Positive	46	Bad borrowers correctly predicted as bad

Table 8: Confusion Matrix Interpretation

False Negatives = **44**. These are borrowers who were actually bad credit risks but the model predicted them as good. In lending, this is dangerous because it means: The bank may approve loans to borrowers who are likely to default.

From *Table 8* we see, the key issue is the recall for bad borrowers. Recall for class 1 = 0.51. This means: Out of all actual bad borrowers, the model catches only 51%.

So, the model is good overall, but from a credit-risk perspective it is not yet conservative enough.

Class	Precision	Recall	F1-score	Meaning
0: Good risk	0.81	0.88	0.84	Model identifies good borrowers well
1: Bad risk	0.65	0.51	0.57	Model struggles more with bad borrowers

Table 9: Classification Report Interpretation

ROC-AUC is: 0.796. So our model is close to the strong range. The ROC curve in *Figure 1* (See Appendix) being clearly above the diagonal random-classifier line means:

Logistic Regression is learning meaningful patterns from the borrower characteristics.

Feature	Coefficient	Interpretation
Status_Checking_Account_A14	-0.786	Lowers probability of bad risk
Credit_History_A34	-0.536	Lowers probability of bad risk
Employment_Since_A74	-0.473	Lowers probability of bad risk
Savings_Account_A65	-0.462	Lowers probability of bad risk
Property_A124	0.440	Raises probability of bad risk
Duration_Months	0.366	Longer loan duration raises probability of bad risk

Installment_Rate	0.363	Higher installment burden raises probability of bad risk
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Table 10: Top coefficients from Coefficient Table

Positive coefficient means: Feature increases probability of bad credit risk

Negative coefficient means: Feature decreases probability of bad credit risk

So, for example:

Duration_Months = positive coefficient means: Longer loan durations are associated with a higher probability of bad credit risk. That makes intuitive sense because longer loans expose the lender to more uncertainty.

Class	Precision	Recall	F1 Score	Support
Good Risk (0)	0.81	0.88	0.84	210
Bad Risk (1)	0.65	0.51	0.57	90
Accuracy			0.77	300
Macro Average	0.73	0.70	0.71	300
Weighted Average	0.76	0.77	0.76	300

Table 11: Classification Report results from Jupyter Notebook

Some features have coded names like: A14, A34, A74, A65. These come from the German Credit dataset's original coding system. For example, Status_Checking_Account_A14 means one particular checking-account category. We should not over-interpret these until we map the codes to their original meanings. For now, it is enough to understand: Negative coefficients reduce bad-risk probability, while positive coefficients increase bad-risk probability.

Interpretation

The Logistic Regression model was used to classify borrowers into good and bad credit risk categories using the German Credit dataset. The target variable was defined as 1 for bad credit risk and 0 for good credit risk.

The model achieved an accuracy of 0.77 on the test set, meaning that it correctly classified 77% of the observations. The ROC-AUC score was 0.796, indicating that the model has a reasonable ability to distinguish between good and bad credit risks.

The confusion matrix shows that the model correctly classified 185 good borrowers and 46 bad borrowers. However, it also misclassified 44 bad borrowers as good borrowers. This is an important limitation because, in credit risk, approving a bad borrower can be more costly than rejecting a good borrower.

The classification report shows that the model performs better for good borrowers than for bad borrowers. The recall for good borrowers is 0.88, while the recall for bad borrowers is only 0.51. This means that the model identifies most good borrowers correctly but catches only about 51% of actual bad borrowers.

The coefficient table shows which variables have the strongest influence on the predicted probability of bad credit risk. Since the target variable is bad credit risk, positive coefficients increase the probability of bad credit risk, while negative coefficients reduce it. For example, loan duration has a positive coefficient, suggesting that longer loan durations are associated with higher credit risk.

Overall, the Logistic Regression model provides a useful first classification model for credit risk. However, the relatively low recall for bad borrowers suggests that the model may need threshold adjustment or cost-sensitive evaluation before being used in a real lending context.

The next useful improvement could be to adjust the classification threshold from 0.50 to something lower, such as 0.40 or 0.30, to catch more bad borrowers. This is especially important in credit risk.

Conclusion

This exercise demonstrates how Logistic Regression can be used for a binary credit-risk classification problem. This problem provides a foundation for more advanced

classification exercises, including threshold tuning and cost-sensitive credit-risk modelling.

Problem 5A: Threshold Tuning and Cost-Sensitive Credit Risk Classification

Research Anchor: Beninel, Bouaguel and Belmufti (2012), “Transfer Learning Using Logistic Regression in Credit Scoring”

Additional Conceptual Anchor: Shen, F., Zhao, X., Kou, G. and Alsaadi, F.E. (2019) ‘A cost-sensitive logistic regression credit scoring model for microfinance institutions’, *Technological and Economic Development of Economy*, 25(3), pp. 365–383.

This paper is used as an additional conceptual anchor because it explicitly focuses on cost-sensitive logistic regression for credit scoring. It argues that ordinary classification models often treat all errors as equally costly, whereas credit scoring requires special attention to the higher cost of incorrectly approving risky borrowers.

It is an extension of Problem 5 using the same German Credit model.

Can we reduce the number of bad borrowers wrongly classified as good borrowers by changing the classification threshold?

In credit risk, not all mistakes are equally costly. We will assume:

- False positive cost = 1: Good borrower wrongly rejected
- False negative cost = 5: Bad borrower wrongly approved

At the default threshold of 0.50, our model had:

Threshold	False Positives	False Negatives	Recall Bad Risk	Accuracy	Total Cost
0.50	25	44	0.5111	0.7700	245

At the cost-sensitive threshold of 0.20, our model has:

Threshold	False Positives	False Negatives	Recall Bad Risk	Accuracy	Total Cost
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0.20	84	14	0.8444	0.6733	154
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So, the threshold-tuning has worked. The model becomes more conservative:

- Lower threshold → catches more bad borrowers
- Lower threshold → rejects more good borrowers wrongly

This is exactly the trade-off we wanted to study.

Missing a bad borrower is five times more costly than wrongly rejecting a good borrower.

At threshold 0.50:

$$\text{Total cost} = 25 \times 1 + 44 \times 5 = 245$$

At threshold 0.20:

$$\text{Total cost} = 84 \times 1 + 14 \times 5 = 154$$

So even though threshold 0.20 creates more false positives, it greatly reduces false negatives, and therefore the total cost falls from **245 to 154**. That is why threshold 0.20 is preferred in this cost-sensitive setting.

Interpretation of Threshold Tuning Results

This extension examines whether changing the classification threshold improves the credit-risk usefulness of the Logistic Regression model. In the baseline model, a borrower was classified as bad risk if the predicted probability of bad credit risk was at least 0.50. However, in credit-risk applications, missing a bad borrower can be more costly than wrongly rejecting a good borrower.

At the default threshold of 0.50, the model produced 44 false negatives, meaning that 44 actual bad borrowers were incorrectly classified as good borrowers. The recall for bad borrowers was 0.5111, meaning that the model identified only about 51% of actual bad-risk borrowers.

After testing multiple thresholds, the cost-sensitive evaluation selected a threshold of 0.20. At this threshold, false negatives fell from 44 to 14, and recall for bad borrowers increased from 0.5111 to 0.8444. This means that the model became much better at identifying risky borrowers.

The trade-off is that false positives increased from 25 to 84, meaning that more good borrowers were wrongly classified as bad risk. Accuracy also decreased from 0.7700 to 0.6733. However, under the assumed cost structure where a false negative is five times more costly than a false positive, the total misclassification cost decreased from 245 to 154.

Overall, the threshold-tuning exercise shows that the default 0.50 threshold is not necessarily optimal in credit-risk modelling. A lower threshold can be more appropriate when the objective is to reduce the costly mistake of approving bad borrowers.

Conclusion

Problem 5A shows that classification thresholds play an important role in credit-risk modelling. While the default threshold of 0.50 gives higher accuracy, it misses too many bad borrowers. A lower threshold of 0.20 reduces false negatives substantially and lowers the total cost under the assumed cost structure. This demonstrates that model evaluation should consider business costs and risk priorities, not just accuracy.

6. Predicting U.S. Recession Probability Using the Yield Curve

Problem 6: Medium Logistic Regression – Predicting U.S. Recession Probability Using the Yield Curve

Research Anchor: Estrella, A. and Mishkin, F.S. (1996) *The Yield Curve as a Predictor of U.S. Recessions*, Federal Reserve Bank of New York, *Current Issues in Economics and Finance*, 2(7).

Additional Institutional Anchor: Federal Reserve Bank of New York, *The Yield Curve as a Leading Indicator*.

The main research anchor studies how the yield-curve spread, especially the difference between long-term and short-term Treasury rates, can predict U.S. recessions.

The New York Fed explains that its model uses the term spread, defined as the difference between 10-year and 3-month Treasury rates, to estimate the probability of a U.S. recession twelve months ahead.

Dataset 1: Federal Reserve Bank of St. Louis, FRED series **T10Y3MM** — *10-Year Treasury Constant Maturity Minus 3-Month Treasury Constant Maturity*.

This gives the monthly term spread that we will use as the predictor. FRED notes that the series is calculated as the spread between the 10-year Treasury constant maturity yield and the 3-month Treasury constant maturity yield.

Dataset 2: Federal Reserve Bank of St. Louis, FRED series **USREC** — *NBER based Recession Indicators for the United States from the Period following the Peak through the Trough*.

This gives the recession indicator, where 1 means recession and 0 means no recession.

In this problem, we use Logistic Regression to estimate whether the U.S. economy is likely to enter a recession within the next 12 months based on the yield-curve spread.

The key idea is:

- Predictor: 10-year Treasury yield minus 3-month Treasury yield
- Target: Recession within the next 12 months

When the yield curve becomes very flat or negative, it may signal that markets expect weaker future economic activity. Logistic Regression is suitable here because the target is binary:

- 0 = No recession within next 12 months
- 1 = Recession within next 12 months

Yield_Spread means: 10-year Treasury yield - 3-month Treasury yield

If this value is negative, the yield curve is inverted.

$$X = \text{yield spread}$$

$$y = \text{recession within next 12 months}$$

Because the predictor is Yield_Spread, the expected sign is usually negative.

That means: Lower yield spread $\rightarrow$ higher recession probability

Higher yield spread $\rightarrow$ lower recession probability

Findings

Metric	Value
Accuracy	0.7383
ROC-AUC	0.5596
Yield Spread Coefficient	-1.0731

Table 12: Metrics from Main Result

The coefficient is the most economically sensible part:

$$\text{Yield Spread coefficient} = -1.0731$$

This means: As the yield spread decreases, the predicted probability of recession within the next 12 months increases. That is exactly what we expected from the yield-curve recession literature.

Actual / Predicted	Predicted No Recession	Predicted Recession
Actual No Recession	178	36
Actual Recession	31	11

Result	Count	Meaning
TN	178	Correctly predicted no recession
FP	36	Predicted recession, but no recession occurred
FN	31	Missed recession cases
TP	11	Correctly predicted recession cases

Table 13: *Confusion Matrix*

The issue is here:

- False Negatives = 31
- True Positives = 11

The model catches only 11 out of 42 recession-within-12-month observations.

For class 0, no recession: The model performs well for non-recession observations.

Metric	Value
Precision	0.85
Recall	0.83
F1-score	0.84

Table 14: *Classification Report for Class 0*

For class 1, recession within next 12 months: This is weak.

Metric	Value
Precision	0.23
Recall	0.26
F1-score	0.25

Table 15: Classification Report for Class 1

The recall for recession is: 0.26

This means: Out of all actual recession-within-12-month observations, the model catches only 26%. So, the model is much better at identifying normal periods than recession-risk periods.

Accuracy is: 0.7383

At first, this looks acceptable. But the test set has many more non-recession observations than recession observations:

- No recession class = 214
- Recession class = 42

So, a model can get decent accuracy simply by predicting “no recession” most of the time. That is why ROC-AUC and recall for class 1 matter more here.

The ROC-AUC is: 0.5596

This is only slightly better than random guessing. So, the conclusion is: The model has the correct economic sign, but its out-of-sample classification performance is weak.

The predicted probability plot is interesting. It shows that predicted recession probability rises before some stress periods, especially around: 2006–2008 | 2018–2020 | 2022–2024

This makes economic sense because the yield curve flattened or inverted before those periods. However, the model does not classify recessions very well using the default 0.50 threshold. Some predicted probabilities rise, but the timing and classification are not strong enough to give high recall or high AUC.

This is a key learning point: A model can produce economically sensible probabilities but still perform poorly as a classifier at a fixed threshold.

This problem teaches four things:

1. Logistic Regression can be used for macroeconomic event prediction.
2. The yield spread has the expected negative relationship with recession probability.
3. Accuracy can be misleading when recession observations are rare.
4. A single predictor may not be enough for strong recession classification.

Interpretation

The Logistic Regression model was used to predict whether the U.S. economy would experience a recession within the next 12 months using the 10-year minus 3-month Treasury yield spread as the predictor. The estimated coefficient on the yield spread is -1.0731, which has the expected sign. This means that a lower yield spread is associated with a higher predicted probability of recession.

The model achieved an accuracy of 0.7383 on the test set. However, the classification report shows that this accuracy is mainly driven by the model's ability to identify non-recession periods. For class 0, representing no recession within the next 12 months, the model achieved a recall of 0.83. In contrast, for class 1, representing recession within the next 12 months, the recall was only 0.26. This means that the model identified only 26% of actual recession-risk observations.

The confusion matrix shows that the model correctly identified 178 non-recession observations and 11 recession observations. However, it missed 31 recession observations and incorrectly classified 36 non-recession observations as recession cases. This indicates that the model has limited effectiveness in identifying recession periods using the default classification threshold.

The ROC-AUC score is 0.5596, which is only slightly above random classification. Therefore, although the yield spread has the expected negative relationship with recession probability, the model's out-of-sample classification performance is weak.

This suggests that a single-variable Logistic Regression model may be too simple for reliable recession prediction, especially when recession observations are rare.

Conclusion

Overall, the exercise shows that the yield curve contains useful economic information, but classification performance depends heavily on model specification, threshold choice, and the rarity of recession events.

Because recession recall is weak, we should do **Problem 6A: Threshold tuning for recession prediction**, just like we did for credit risk. That would show whether lowering the classification threshold improves recession detection.

Problem 6A: Threshold tuning for Recession prediction

Research Anchor: Estrella, A. and Mishkin, F.S. (1996) *The Yield Curve as a Predictor of U.S. Recessions*, Federal Reserve Bank of New York, *Current Issues in Economics and Finance*, 2(7).

Dataset 1: FRED series T10Y3MM — *10-Year Treasury Constant Maturity Minus 3-Month Treasury Constant Maturity*.

Dataset 2: FRED series USREC — *NBER based Recession Indicators for the United States from the Period following the Peak through the Trough*.

In Problem 6, the default Logistic Regression threshold was: 0.50

But recession events are rare. So, the default 0.50 threshold may miss too many recession-risk observations.

In our Problem 6 result, recall for recession class was only: 0.26

So Problem 6A asks: Can we improve recession detection by lowering the classification threshold?

Metric	Default Threshold (0.50)	Tuned Threshold (0.35)
True Negatives	179	148
False Positives	36	67
False Negatives	31	19
True Positives	11	23
Accuracy	0.7393	0.6654
Precision Recession	0.2340	0.2556
Recall Recession	0.2619	0.5476
F1 Recession	0.2472	0.3485
Total Cost	191	162

Table 16: Default v. Tuned Thresholds

At the default threshold of 0.50, the model caught only 11 recession-risk observations and missed 31 recession-risk observations.

After lowering the threshold to 0.35, the model caught 23 recession-risk observations and missed only 19 recession-risk observations.

So, the recall for recession improved from 0.2619 to 0.5476. This means that the model became **much better at detecting recession-risk periods**.

The trade-off is that false positives increased from 36 to 67. This means the model gives more recession warnings when no recession occurs.

Accuracy also fell from 0.7393 to 0.6654. But this is expected. In recession prediction, accuracy is not always the best objective because recessions are rare. A model can get high accuracy simply by predicting “no recession” most of the time.

So, the real question is: Are we willing to accept more false alarms in order to catch more recession-risk periods? In this cost-sensitive setup, the answer is yes.

We assumed:

- False positive cost = 1
- False negative cost = 5

So, missing a recession-risk period is treated as five times more costly than giving a false recession warning.

- At threshold 0.50, the Total cost = $36 \times 1 + 31 \times 5 = 191$
- At threshold 0.35, the Total cost = $67 \times 1 + 19 \times 5 = 162$

So, the tuned threshold reduces total cost from 191 $\rightarrow$ 162. That is why threshold 0.35 is preferred.

Interpretation

Problem 6A extends the yield-curve recession prediction model by examining whether the default Logistic Regression threshold of 0.50 is appropriate for recession

classification. Since recessions are relatively rare, a model may achieve reasonable accuracy while still missing many recession-risk observations. Therefore, this extension evaluates different classification thresholds and compares their effects on recession recall, precision, F1-score, and total misclassification cost.

At the default threshold of 0.50, the model achieved an accuracy of 0.7393. However, it identified only 11 recession-risk observations and missed 31. The recall for the recession class was only 0.2619, meaning that the model detected about 26% of actual recession-risk observations.

After testing multiple thresholds, **the best cost-sensitive threshold was 0.35**. At this threshold, the model identified 23 recession-risk observations and reduced false negatives from 31 to 19. Recession recall improved from 0.2619 to 0.5476, and the F1-score for the recession class improved from 0.2472 to 0.3485.

The **trade-off is that false positives increased from 36 to 67**, and overall accuracy declined from 0.7393 to 0.6654. This means the model became more sensitive to recession risk but produced more false alarms. Under the assumed cost structure, where missing a recession signal is five times more costly than giving a false warning, the total cost decreased from 191 to 162.

Overall, the threshold-tuning exercise shows that the default 0.50 threshold is not necessarily suitable for recession prediction. A lower threshold of 0.35 provides a better balance when the objective is to detect recession risk rather than maximize overall accuracy.

Conclusion

Problem 6A shows that **classification threshold choice is important in macroeconomic event prediction**. The default threshold of 0.50 produced higher accuracy but missed too many recession-risk observations. Lowering the threshold to 0.35 improved recession recall and reduced the total cost under the assumed cost-sensitive framework. This demonstrates that for rare but important events such as recessions, model evaluation should focus on recall, false negatives, and cost-sensitive performance rather than accuracy alone.

7. Forecasting Monthly Stock-Market Direction

Problem 7: Hard Logistic Regression - Forecasting Monthly Stock-Market Direction

Research Anchor: Nyberg, H. (2011) ‘Forecasting the direction of the US stock market with dynamic binary probit models’, *International Journal of Forecasting*, 27(2), pp. 561–578.

This research studies whether the sign of monthly excess stock returns can be predicted using binary-choice models. Nyberg’s paper uses dynamic binary probit models, while our beginner-friendly replication-style version will use Logistic Regression.

Dataset 1: Kenneth R. French Data Library — **Fama/French 3 Factors**. We will use the monthly market excess return, Mkt-RF, as the stock-market excess return series. The Kenneth French Data Library provides downloadable factor and portfolio return datasets widely used in empirical asset-pricing research.

Dataset 2: FRED series **T10Y3MM** — *10-Year Treasury Constant Maturity Minus 3-Month Treasury Constant Maturity*. We will use this as a macro-financial predictor because it measures the Treasury yield spread between long-term and short-term interest rates.

Dataset 3: FRED series **USREC** — *NBER based Recession Indicators for the United States from the Period following the Peak through the Trough*. We will use this as a business-cycle indicator. FRED describes this series as an NBER-based recession indicator for the United States.

We use Logistic Regression to predict whether the **next month’s U.S. market excess return** will be positive or negative.

The target variable is:

- 1 = next month market excess return is positive
- 0 = next month market excess return is zero or negative

This is harder than the previous Logistic Regression problems because stock-market direction is noisy and difficult to forecast. The model is not trying to predict the exact return. It is only trying to predict the **direction**. So, the research question is:

Can lagged market information and macro-financial variables help predict whether the next month's stock-market excess return will be positive?

Metric	Value
Accuracy	0.6070
ROC-AUC	0.5213

Table 17: Metrics from Main Result

At first, accuracy of 60.7% may look acceptable. But we need to compare it with a very simple benchmark.

Our test set has:

Class	Meaning	Count
0	Market down or zero next month	92
1	Market up next month	165

So, if we simply predicted **“market up” every month**, the benchmark accuracy would be: $165/257 = 0.642$

That means a naive “always predict up” model would get about **64.2% accuracy**, which is **better than your Logistic Regression model's 60.7% accuracy**. So, the model does **not** beat the simple benchmark.

Actual / Predicted	Predicted Down	Predicted Up
Actual Down	0	92
Actual Up	9	156

Result	Count	Meaning
True negatives	0	The model correctly predicted zero down months
False positives	92	The model predicted up when the market was actually down
False negatives	9	The model predicted down when the market was actually up
True positives	156	The model correctly predicted many up months

Table 18: *Confusion Matrix*

The biggest problem is: True negatives = 0

The model completely fails to identify negative market-return months.

It is mostly predicting: Market will go up next month

That is why the recall for class 1 is high, but the recall for class 0 is zero.

Metric	Value
Precision	0.63
Recall	0.95
F1-score	0.76

Table 19: *Classification Table for Class 1 (Market Up)*

Metric	Value
Precision	0.00
Recall	0.00
F1-score	0.00

Table 20: *Classification Table for Class 0 (Market Down)*

The real weakness is visible in *Table 18*. The model has no ability to identify down months at the 0.50 threshold.

Our ROC-AUC is: 0.5213. This is only slightly above 0.50. Since 0.50 means random guessing, this result means:

The model has almost no useful ability to separate positive-return months from negative-return months. This is a realistic outcome. Stock-market direction is extremely difficult to forecast using simple lagged factors.

Feature	Coefficient
HML_Lag1	-0.1410
Mkt_RF_Lag1	-0.1045
Rolling_12M_Volatility	-0.0715
Recession_Lag1	-0.0492
SMB_Lag1	-0.0146
Yield_Spread_Lag1	-0.0053

Table 21: Coefficient Table

All coefficients are negative. Since the target is: 1 = market up next month, negative coefficients mean:

Higher values of these predictors are associated with a lower predicted probability of a positive market return next month.

But because the model's forecasting performance is weak, we should **not over-interpret** these coefficients. The coefficients exist mathematically, but the model does not produce strong predictive power.

The plot shows that predicted probabilities are mostly above 0.50, often around: 0.55 to 0.65

This explains why the model predicts “up” almost every month. The model rarely gives probabilities below 0.50, so it rarely predicts a down month. That is why the confusion matrix shows zero correctly predicted down months.

This problem teaches an important lesson. Logistic Regression can be applied to stock-market direction prediction, but that does not mean it will produce useful forecasting performance.

The model is **technically correct, but empirically weak.**

This is exactly why stock-market prediction is considered hard. Financial returns are noisy, competitive, and often close to unpredictable using simple public predictors.

Interpretation

The Logistic Regression model was used to forecast whether the next month's U.S. market excess return would be positive or non-positive. The target variable was defined as 1 if the next month's market excess return was positive and 0 otherwise. The predictors included lagged market excess return, SMB, HML, the yield spread, recession indicator, and rolling 12-month market volatility.

The model achieved an accuracy of 0.6070 and an ROC-AUC score of 0.5213. Although the accuracy appears to be above 60%, the ROC-AUC score is only slightly above 0.50, which indicates that the model has very weak ability to distinguish between positive and non-positive market-return months. Since a ROC-AUC score of 0.50 represents random classification, the result suggests that the model provides little useful directional forecasting power.

The confusion matrix shows that the model correctly predicted 156 positive-return months but failed to correctly identify any non-positive-return months. Specifically, there were 0 true negatives and 92 false positives. This means that the model predicted the market would go up in almost all cases, even when the market actually went down or produced a non-positive excess return.

The classification report confirms this weakness. For class 1, representing a positive market excess return, the model achieved high recall of 0.95. However, this is mainly because the model predicted positive returns most of the time. For class 0, representing a non-positive market excess return, precision, recall, and F1-score were all 0.00. This shows that the model was unable to identify negative or zero-return months.

The coefficient table shows that all predictors had negative coefficients, including lagged HML, lagged market excess return, rolling volatility, recession indicator, SMB, and the yield spread. Since the target variable is a positive market return next month, negative coefficients imply that higher values of these predictors are associated with a lower predicted probability of a positive return. However, because the model's predictive performance is weak, these coefficients should not be over-interpreted.

The predicted probability plot shows that the model's estimated probabilities were mostly above 0.50, generally clustering between 0.55 and 0.65. This explains why the model predicted positive market returns in most months. The model rarely assigned probabilities below the default threshold of 0.50, which caused it to miss almost all non-positive-return months.

Conclusion

Overall, the results suggest that this Logistic Regression specification is not effective for forecasting monthly stock-market direction. The model does not meaningfully outperform random classification and fails to identify down-market months. This is an important empirical result because stock-market direction is known to be difficult to predict using simple lagged return and macro-financial variables. The exercise demonstrates that a model can be technically valid and still have limited practical forecasting value.

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FRED series USREC — *NBER based Recession Indicators for the United States from the Period following the Peak through the Trough*.

Appendices

A.5 Logistic Regression – Credit Risk Classification Using German Credit Data

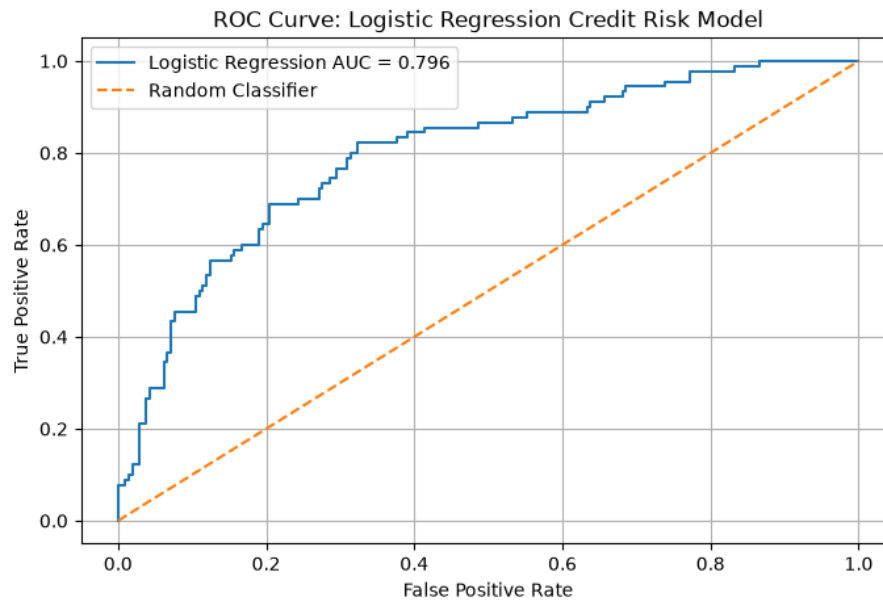


Figure 1: ROC Curve for Logistic Regression Credit Risk Classification

Note: The ROC curve shows the trade-off between the true positive rate and false positive rate across different classification thresholds. The curve suggests that the Logistic Regression model has reasonable discriminatory power in distinguishing between good and bad credit risks.

A.5A Threshold Tuning and Cost-Sensitive Credit Risk Classification

	Threshold	True Negatives	False Positives	False Negatives	True Positives	Accuracy	Precision Bad Risk	Recall Bad Risk	F1 Bad Risk
0	0.10	74	136	8	82	0.520000	0.376147	0.911111	0.532468
1	0.15	108	102	13	77	0.616667	0.430168	0.855556	0.572491
2	0.20	126	84	14	76	0.673333	0.475000	0.844444	0.608000
3	0.25	143	67	18	72	0.716667	0.517986	0.800000	0.628821
4	0.30	151	59	24	66	0.723333	0.528000	0.733333	0.613953
5	0.35	162	48	28	62	0.746667	0.563636	0.688889	0.620000
6	0.40	170	40	34	56	0.753333	0.583333	0.622222	0.602151
7	0.45	175	35	37	53	0.760000	0.602273	0.588889	0.595506
8	0.50	185	25	44	46	0.770000	0.647887	0.511111	0.571429
9	0.55	193	17	49	41	0.780000	0.706897	0.455556	0.554054
10	0.60	195	15	55	35	0.766667	0.700000	0.388889	0.500000
11	0.65	197	13	63	27	0.746667	0.675000	0.300000	0.415385
12	0.70	202	8	68	22	0.746667	0.733333	0.244444	0.366667
13	0.75	205	5	79	11	0.720000	0.687500	0.122222	0.207547
14	0.80	207	3	81	9	0.720000	0.750000	0.100000	0.176471
15	0.85	210	0	83	7	0.723333	1.000000	0.077778	0.144330
16	0.90	210	0	88	2	0.706667	1.000000	0.022222	0.043478

Figure 2: Threshold Results

	Threshold	True Negatives	False Positives	False Negatives	True Positives	Accuracy	Precision	Bad Risk	Recall	Bad Risk	F1	Bad Risk	Total Cost
0	0.10	74	136	8	82	0.520000		0.376147		0.911111		0.532468	176
1	0.15	108	102	13	77	0.616667		0.430168		0.855556		0.572491	167
2	0.20	126	84	14	76	0.673333		0.475000		0.844444		0.608000	154
3	0.25	143	67	18	72	0.716667		0.517986		0.800000		0.628821	157
4	0.30	151	59	24	66	0.723333		0.528000		0.733333		0.613953	179
5	0.35	162	48	28	62	0.746667		0.563636		0.688889		0.620000	188
6	0.40	170	40	34	56	0.753333		0.583333		0.622222		0.602151	210
7	0.45	175	35	37	53	0.760000		0.602273		0.588889		0.595506	220
8	0.50	185	25	44	46	0.770000		0.647887		0.511111		0.571429	245
9	0.55	193	17	49	41	0.780000		0.706897		0.455556		0.554054	262
10	0.60	195	15	55	35	0.766667		0.700000		0.388889		0.500000	290
11	0.65	197	13	63	27	0.746667		0.675000		0.300000		0.415385	328
12	0.70	202	8	68	22	0.746667		0.733333		0.244444		0.366667	348
13	0.75	205	5	79	11	0.720000		0.687500		0.122222		0.207547	400
14	0.80	207	3	81	9	0.720000		0.750000		0.100000		0.176471	408
15	0.85	210	0	83	7	0.723333		1.000000		0.077778		0.144330	415
16	0.90	210	0	88	2	0.706667		1.000000		0.022222		0.043478	440

Figure 3: Cost Sensitive Evaluation for Credit Risk Classification

Threshold	0.200000
True Negatives	126.000000
False Positives	84.000000
False Negatives	14.000000
True Positives	76.000000
Accuracy	0.673333
Precision Bad Risk	0.475000
Recall Bad Risk	0.844444
F1 Bad Risk	0.608000
Total Cost	154.000000

Figure 4a: Best Cost Threshold

Threshold	0.250000
True Negatives	143.000000
False Positives	67.000000
False Negatives	18.000000
True Positives	72.000000
Accuracy	0.716667
Precision Bad Risk	0.517986
Recall Bad Risk	0.800000
F1 Bad Risk	0.628821
Total Cost	157.000000

Figure 4b: Best F1 Threshold

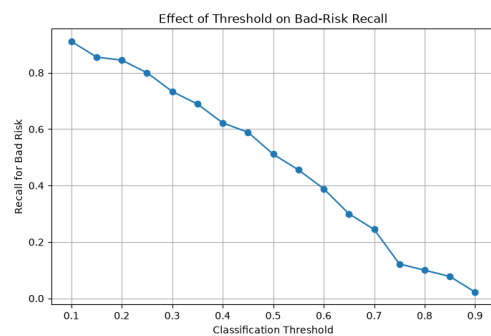


Figure 5a: Threshold and Bad Risk Recall

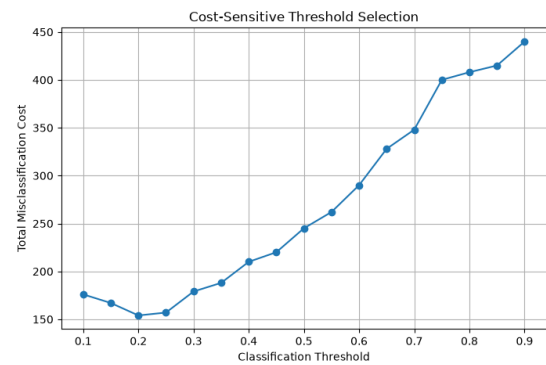


Figure 5b: Cost Sensitive Threshold

A.6 Predicting U.S. Recession Probability Using the Yield Curve

	precision	recall	f1-score	support
0	0.85	0.83	0.84	214
1	0.23	0.26	0.25	42
accuracy			0.74	256
macro avg	0.54	0.55	0.54	256
weighted avg	0.75	0.74	0.74	256

Figure 6: Classification Report for US Recession

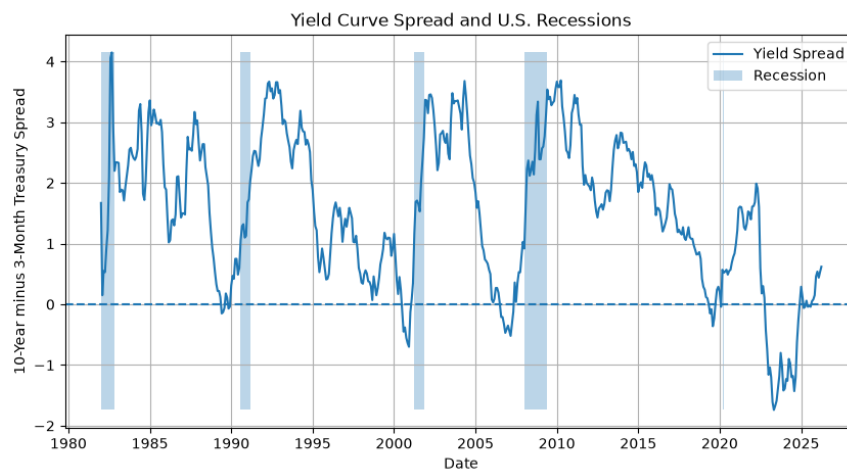


Figure 7: Yield Curve Spread and US Recessions

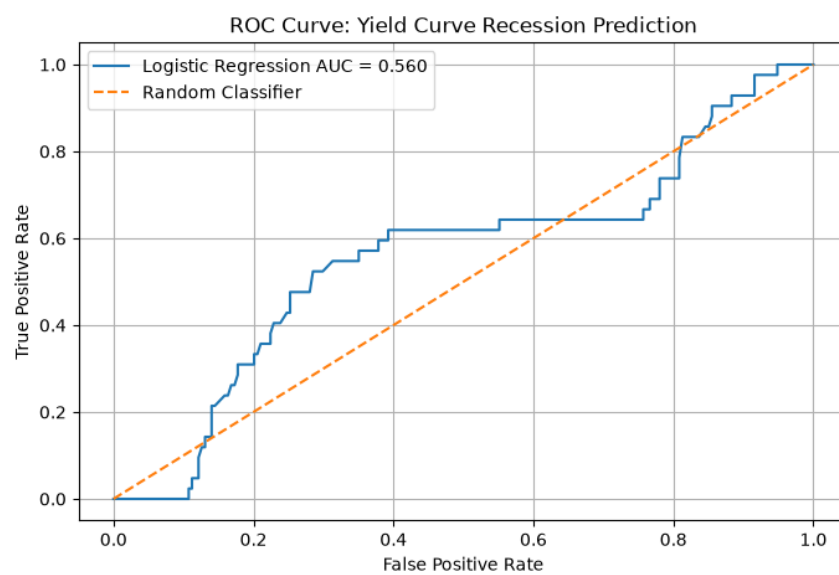


Figure 8: ROC Curve for Yield Recession Prediction

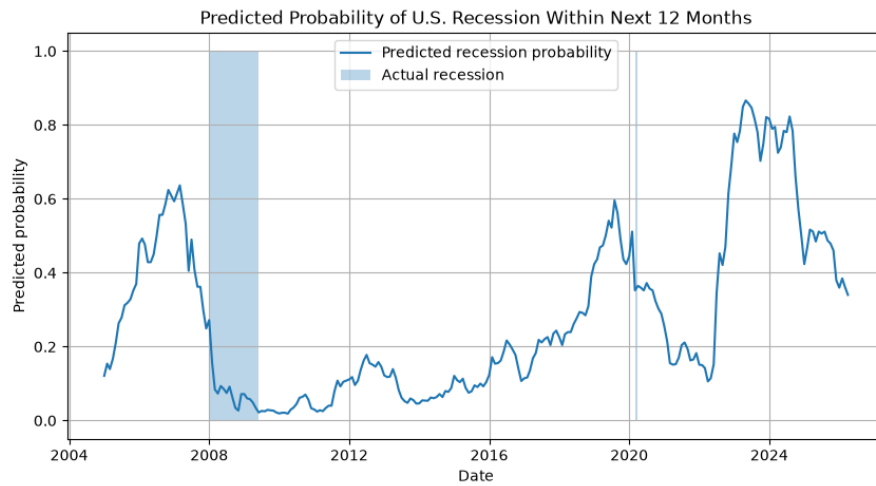


Figure 9: Predicted Probability of US Recession

A.6A Threshold tuning for Recession Prediction

	Threshold	True Negatives	False Positives	False Negatives	True Positives	Accuracy	Precision Recession	Recall Recession	F1 Recession	Total Cost
0	0.05	25	190	4	38	0.245136	0.166667	0.904762	0.281481	210
1	0.10	54	161	15	27	0.315175	0.143617	0.642857	0.234783	236
2	0.15	86	129	15	27	0.439689	0.173077	0.642857	0.272727	204
3	0.20	112	103	16	26	0.536965	0.201550	0.619048	0.304094	183
4	0.25	130	85	17	25	0.603113	0.227273	0.595238	0.328947	170
5	0.30	139	76	19	23	0.630350	0.232323	0.547619	0.326241	171
6	0.35	148	67	19	23	0.665370	0.255556	0.547619	0.348485	162
7	0.40	161	54	22	20	0.704280	0.270270	0.476190	0.344828	164
8	0.45	168	47	27	15	0.712062	0.241935	0.357143	0.288462	182
9	0.50	179	36	31	11	0.739300	0.234043	0.261905	0.247191	191
10	0.55	185	30	36	6	0.743191	0.166667	0.142857	0.153846	210
11	0.60	189	26	40	2	0.743191	0.071429	0.047619	0.057143	226
12	0.65	192	23	42	0	0.747082	0.000000	0.000000	0.000000	233
13	0.70	194	21	42	0	0.754864	0.000000	0.000000	0.000000	231
14	0.75	198	17	42	0	0.770428	0.000000	0.000000	0.000000	227
15	0.80	207	8	42	0	0.805447	0.000000	0.000000	0.000000	218
16	0.85	213	2	42	0	0.828794	0.000000	0.000000	0.000000	212
17	0.90	215	0	42	0	0.836576	0.000000	0.000000	0.000000	210
18	0.95	215	0	42	0	0.836576	0.000000	0.000000	0.000000	210

Figure 10: Threshold Results with Cost Sensitive Evaluation

```

Threshold          0.350000
True Negatives     148.000000
False Positives    67.000000
False Negatives    19.000000
True Positives     23.000000
Accuracy           0.665370
Precision Recession 0.255556
Recall Recession   0.547619
F1 Recession       0.348485
Total Cost         162.000000
Name: 6, dtype: float64
    
```

Figure 11a: Best Cost Threshold

```

Threshold          0.350000
True Negatives     148.000000
False Positives    67.000000
False Negatives    19.000000
True Positives     23.000000
Accuracy           0.665370
Precision Recession 0.255556
Recall Recession   0.547619
F1 Recession       0.348485
Total Cost         162.000000
Name: 6, dtype: float64
    
```

Figure 11b: Best F1 Threshold

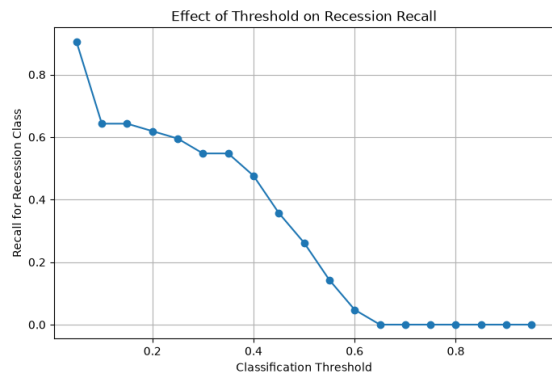


Figure 12a: Threshold v. Recession Recall

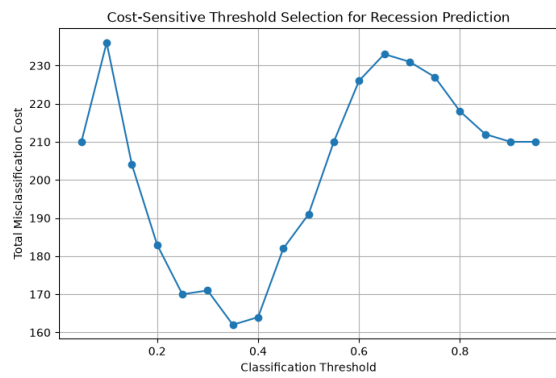


Figure 12b: Threshold v. Total Cost

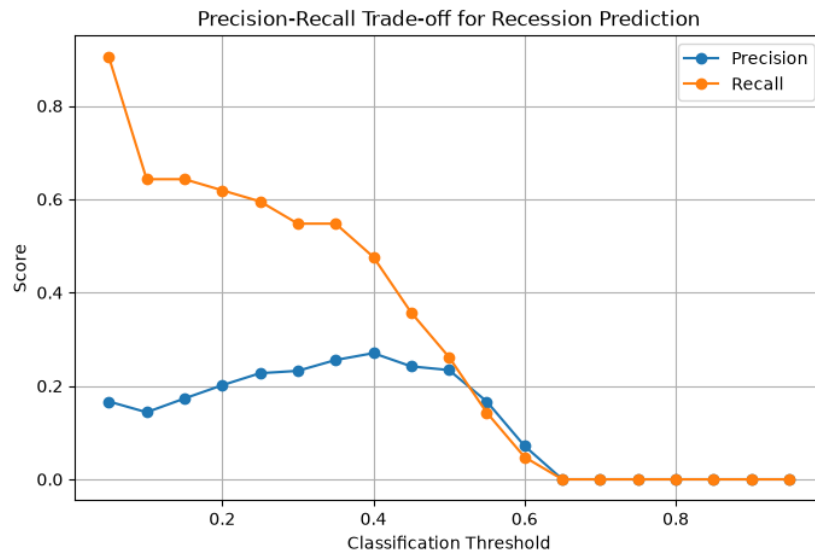


Figure 13: Precision - Recall Trade-off

A.7 Forecasting Monthly Stock-Market Direction

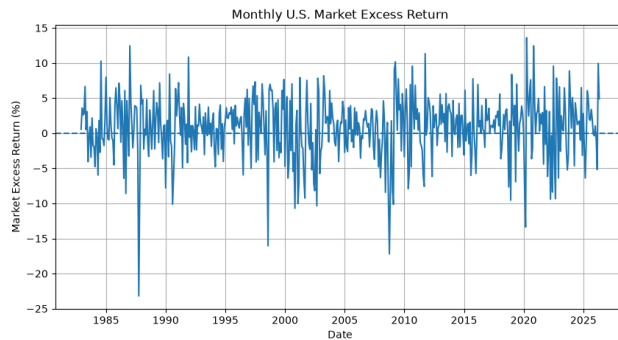


Figure 14a: Market Excess Return over Time

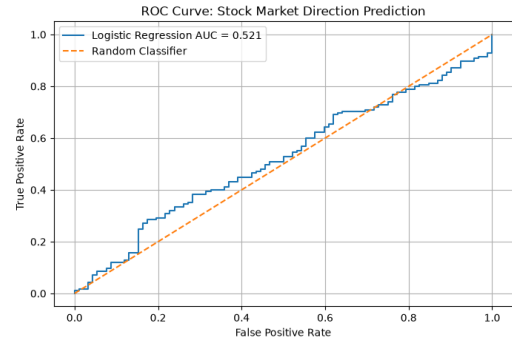


Figure 14b: ROC Curve

	Date	Next_Month_Mkt_RF	Market_Up_Next_Month	Predicted_Probability_Up	Predicted_Class
276	2005-01-01	1.88	1	0.640260	1
277	2005-02-01	-1.94	0	0.620937	1
278	2005-03-01	-2.61	0	0.634975	1
279	2005-04-01	3.65	1	0.661928	1
280	2005-05-01	0.58	1	0.626866	1

Figure 15: Prediction Table for Stock Market Direction

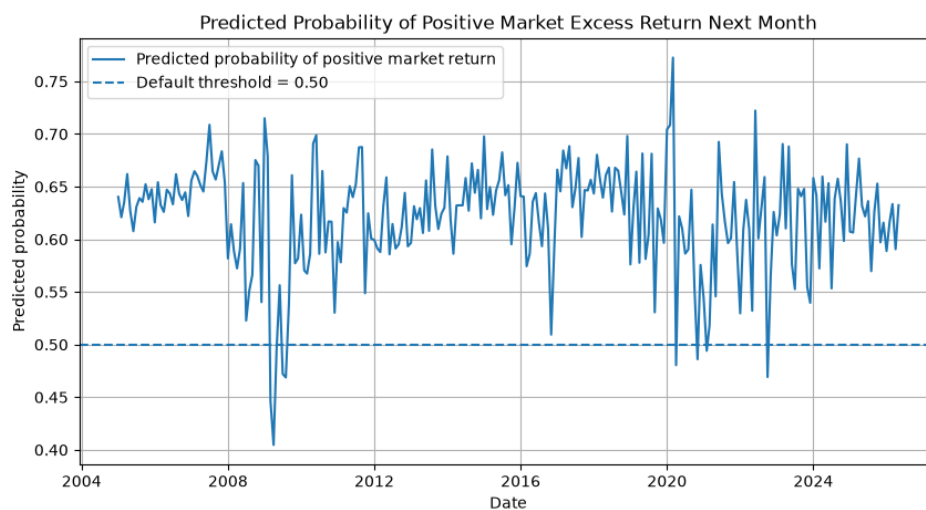


Figure 16: Predicted Probability of Market Up